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Multivectorial magnetization in late Palaeozoic volcanics from North Sardinia; partial remagnetization and rotation

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Summary. Detailed thermal demagnetization results combined with vector analysis and study of the convergence point of remagnetization circles reveal that the late Palaeozoic ignimbrites of North Sardinia possess a multi component remanence in addition to having experienced a tectonic rotation. The degree of palaeomagnetic complexity increases with increasing degree of oxidation of the magnetic mineralogy. It is concluded that the rocks were laid down in late Permian time just before the close of the reversed Kiaman geomagnetic epoch. Subsequent oxidation and partial remagnetization basically occurred in late Permian–Triassic time, during a period characterized by alternating field polarity. In the majority of the sites this remagnetization cycle brought about fairly erratic and relatively stable resultant magnetizations which are generally smeared out towards easterly directions. At a later date Sardinia was subject to an anticlockwise rotation of about 45 degrees, after which a minor chemical magnetization, aligned along the direction of the present axial dipole field, seems to have been acquired by some specimens.

1 Introduction

In recent years considerable interest has been focused on possible movements of micro-continental blocks in the West Mediterranean area. Palaeomagnetic data have played a key role in this issue as several independent studies have suggested anticlockwise rotation of Iberia, Sardinia, Corsica and Italy (see for example de Jong *et al.* 1973; Nairn & Westphal 1968; Soffel 1972; Storetvedt 1973a; van der Voo 1969; Zijderveld & de Jong 1969; Zijderveld, de Jong & van der Voo 1970; Westphal, Orsini & Vellutini 1976). However, at least some of the rotation estimates do not seem to have a firm experimental basis. Thus, Storetvedt (1970) noted that investigated Permian formations in the region, though showing dominantly southeasterly magnetizations, tended to have remanence distributions elongated

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in the direction of declination. For example, Permian rocks from Sardinia (Zijderveld *et al.* 1970) have shown a declination spread of nearly 90° while the total inclination variation amounts to only half of this. Such magnetization structures may either (a) be the result of superimposed normal and reversed components, i.e. the observed southeasterly resultant magnetizations have no tectonic explanation, or (b) a certain remagnetization structure (as under (a)) 'overprints' an actually rotated magnetization. In the latter case the true rotation figure may easily be obscured if the remagnetization effect is not eliminated.

The present paper deals with this rotation/remagnetization issue and the rock formation under consideration is the Permian volcanics in the Gallura region of North Sardinia (Traversa 1966; Vardabasso 1959). Short accounts of the bulk palaeomagnetic properties of these rocks have been given by Zijderveld *et al.* (1970) and Westphal *et al.* (1976) but details on the palaeomagnetic build-up and on magnetic mineralogy have not been presented.

2 Palaeomagnetic methods

A total of 37 oriented blocks from 11 sites have been investigated. The distribution of collected sites is shown in Fig. 1. The sampled material which embraces rhyolitic tuffs and a few ignimbritic lavas varies in colour from dark grey to pinkish. The formation sits approximately horizontal (Traversa 1966) and the local structural situation appears to be original. Therefore, no tectonic corrections have been applied for the investigated sites.

On average, seven specimens from each site have been subjected to stepwise thermal demagnetization up to a maximum of 650°C in a residual magnetic field of about 5 γ. For remanence measurements an astatic magnetometer as well as commercial spinner magnetometers have been used after each heating step.

Vector subtraction analysis (*cf.* e.g. Roy & Park 1974) has been carried out on all specimens defining directional paths on progressive demagnetization. By this technique one can determine the direction of an erased vector provided that there is a range of demagnetization for which only this one component is being removed. When directional agreement is found between various difference vector estimates (along a directional path), it is suggested that this basic requirement has been fulfilled. In case of more closely overlapping stability spectra

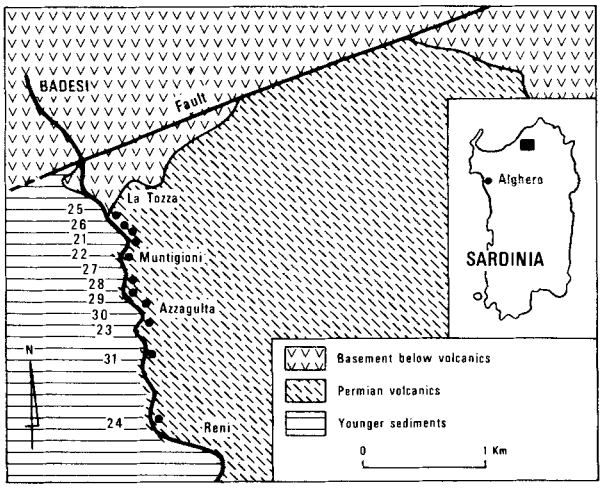


Figure 1. Geological sketch map of Gallura ignimbrites. Simplified after Traversa (1966). Numbered dots are sampling sites which are situated along (or close to) the road between La Tozza and Reni.

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of subcomponents, the subtracted vectors may either define a directional trend or have an erratic distribution. In the latter case vector subtraction results are of course meaningless. For a successful application of this technique it is important to choose demagnetization steps that make the difference vector large enough to be not unduly influenced by random measurement errors. Further details on vector subtraction are given in the discussion of individual results. Converging remagnetization circles (Halls 1976) have been used to estimate the mean direction of a component of secondary magnetization.

3 Some magneto-mineralogical properties

The opaque mineral content is very low in all sites. The characteristic microscopic feature is the occurrence of numerous patches of red silicate groundmass generally interspersed with haematite grains. The larger haematite particles are polycrystalline. Cubic oxides have been detected optically only rarely. When they occur as somewhat larger grains they are always very inhomogeneously magnetized (maghemitization ?) as indicated by the fact that only small and irregularly distributed areas are sufficiently magnetic to attract magnetic colloid. Throughout most of the specimens analysed there are finely dispersed opaques, which have not been optically identified due to their small grains size (~ 1 μ). In sites 23 and 30, which are tuffaceous, the fine-grained opaques (along with the commonly occurring red coloured matrix) form the only observable potential magnetic sources. Sites 23 and 30 appear to have practically identical rock magnetic properties: namely (1) they attain saturation remanence in fields of about 3 kG (Fig. 2b), (2) they exhibit an extremely narrow blocking temperature range at around 550°C (Fig. 2c) and (3) they lose their natural magnetic moment when cooled below about −140°C (Fig. 2a) (*cf.* Nagata & Kobayashi 1964). These observations

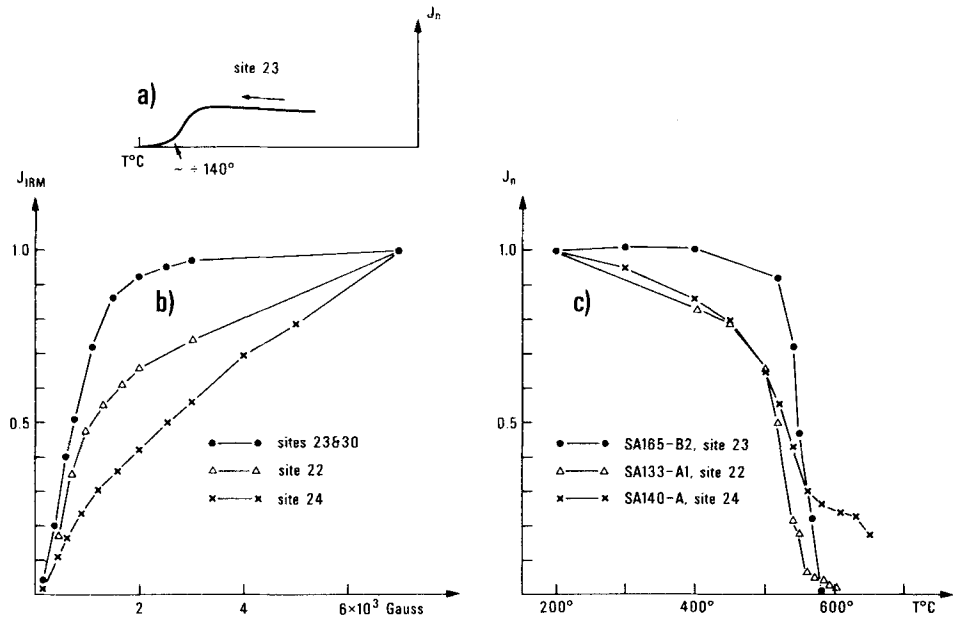


Figure 2. Examples of rock magnetic properties: (a) low-temperature variation of natural intensity, (b) induced remanent magnetization versus increasing magnetizing field and (c) thermal demagnetization of the natural magnetic moment. Further details in text.

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indicate that the finely dispersed opaques are pure magnetite. Also, sites 23 and 30 have nearly reversible J_s–T curves with Curie temperatures (T_c) around 560°C. Based on the thermal decay pattern of the natural remanence (NRM), magnetite also seems to be an important carrier of remanence in the other sites (*cf.* Fig. 2c) though blocking temperatures above 600°C are quite common. In particular, one recognizes the great variability between magnetite (or possibly cation-deficient spinels) and haematite as remanence carriers even within a single hand sample. Haematite is quantitatively a very important magnetic mineral especially in samples from site 24 (*cf.* Figs 2b and c).

On the whole, it appears that the remanence carrying minerals from sites 23 and 30, which provide the most finely grained rocks in this study, are less oxidized (if oxidized at all) than the rest of the rock collection. Also, the natural intensity of magnetization (J_n) for these two sites (varying between 1×10^{-3} and 6×10^{-3} emu/cm^3) is, on average, about two orders of magnitude greater than at the other localities (total variation between 5×10^{-7} and 9×10^{-5} emu/cm^3). This magneto-mineralogical difference is also reflected in the J_s–T analysis: material from sites 23 and 30 have relatively strong saturation moments while on the whole, at the remaining sites, except no. 22, the high field magnetization is too weak to give meaningful Curie points. In the J_s–T analysis 7 kG was the maximum available field and attempts at magnetic separation were unsuccessful.

4 Thermal demagnetization; resolution of superimposed magnetizations

The present study has revealed a considerably more complex palaeomagnetic record than that apparently suggested in the previous studies (Zijderveld *et al.* 1970; Westphal *et al.* 1976). It is imperative therefore to document this case through a reasonable number of examples and in particular to elucidate the remanence build-up within individual samples and sites. This has been attempted in Figs 3–6. Palaeomagnetically there seems to be no general difference between lavas and tuffs.

Site 21, Fig. 3, represents a typical example. The quite stable NRM is fairly scattered with generally shallow inclinations and with declinations displaying a pronounced spread of the order of 100° of arc in the SW and SE quadrants. This complex structure is noted even within a single hand sample. Swinging directional behaviour on progressive thermal demagnetization characterizes the magnetization but stable end points are only rarely achieved in these cases. Vector subtraction analysis is generally not successful for the swinging specimens, implying that to a large extent one is dealing with a system of differently oriented subcomponents having closely overlapping stability spectra. For the whole study the vector subtraction technique appears to have successfully retrieved underlying components of magnetization in fewer then 10 specimens. Fig. 3 shows two of these examples, SA 129-D1 and SA 129-B2. For SA 129-D1 three vector subtracted directions within the temperature range 540–640°C suggest that the direction of the magnetization being erased corresponds to the bulk direction of magnetization between 300 and 540°C. Thus the overall mean direction of vector subtracted directions and directions achieved on progressive demagnetization between 300 and 540°C is likely to give a fair estimate of a true (relative) palaeomagnetic field component. Specimen SA 129-B2 (Fig. 3c) performs a consistent westerly directional trend in the temperature region 200–520° and vector subtraction between various demagnetization steps (within this range of temperature) give direction data consistent with the main magnetization component for specimen SA 129-D1. Above 520°C, specimen SA 129-B2 diverges from its previous directional path, becoming first steeply normal and positive (down) before ending up with a high temperature direction ($T > 600$°C) in the same position as the magnetization component that was erased below 520°C. This

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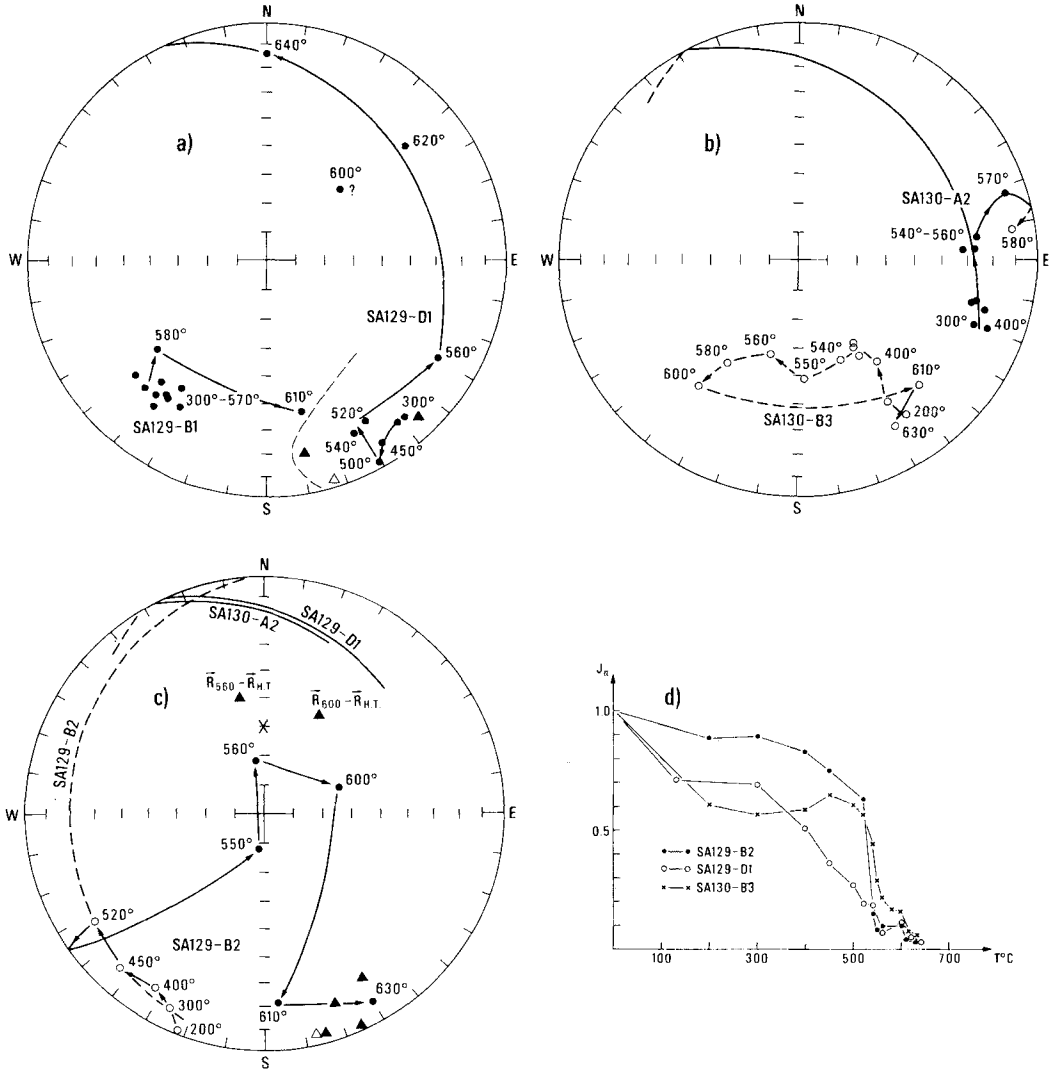


Figure 3. Thermal demagnetization results for specimens from site 21. Sa 129-D1 denotes specimen D1 cut from sample SA 129 etc. Note the within-sample smeared distribution of the fairly stable bulk magnetization and the tendency of changing directional pattern during progressive demagnetization. Intensity diagrams (d) are shown for interesting specimens. Extended best-fitting, northerly trending, great circle paths are shown for three of the specimens (diagram c). Vector subtracted directions (*cf.* Table 1 concerning temperature ranges) are marked by triangles. Open (closed) symbols are upward (downward) pointing magnetization vectors. HT (in diagram c) means 'high temperature' and the cross represents the present axial dipole field direction. Projection is equal area. Further details in text.

directional agreement between the vector subtracted component and the high temperature magnetization reinforces the palaeomagnetic reliability of this SSE direction. Subtraction of the high temperature component (the direction of which is defined by the mean of the 610 and 630°C directions and the intensity that for 610°C) from the 560 and 600°C resultant vectors indicates the existence of a moderately steep, downward dipping, normal component of magnetization. Fig. 3c also shows that the extended remagnetization circles defined by

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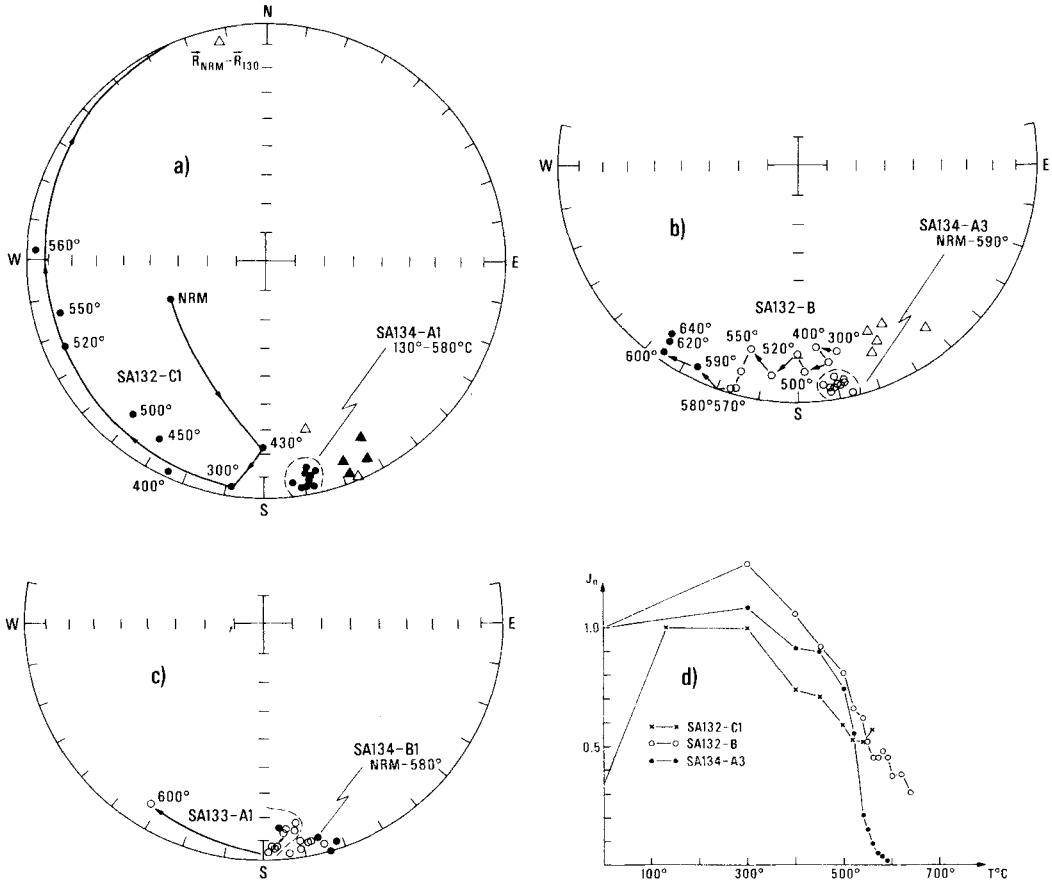


Figure 4. Thermal demagnetization results from site 22. General information as for Fig. 3. Further explanation in text.

the northerly trending paths of specimens SA 129-B2, SA 129-D1 and SA 130-A2 have a convergence 'point' that is roughly anti-parallel to the SSE direction of this site. The best intersection point (Halls 1976) of these and other remagnetization circles will be given in the following chapter.

Site 22, Fig. 4, differs from site 21 in that blocking temperatures above 600°C are much less frequent. Specimens with blocking temperatures confined to the magnetite region exhibit complete directional stability defining practically horizontal, SSE directed magnetization vectors. Examples of this kind are given by specimens SA 134-A1 (Fig. 4a), SA 134-A3 (Fig. 4b) and SA 134-B1 (Fig. 4c). On the other hand, specimens having a relatively strong haematite component show the same swinging behaviour on progressive demagnetization as do the specimens from site 21. Vector subtraction has been successful for two of these latter specimens: SA 132-C1 (Fig. 4a) and SA 132-B (Fig. 4b). For specimen SA 132-C1 subtraction of the 130°C vector from the NRM suggests the existence of a component with low blocking temperature that is nearly anti-parallel to the more stable component, extracted by vector subtraction in the temperature range 300–560°C. Extension of the northward trending remagnetization circle passes close to the inferred low blocking temperature component suggesting that the NNW magnetization of this specimen consists of

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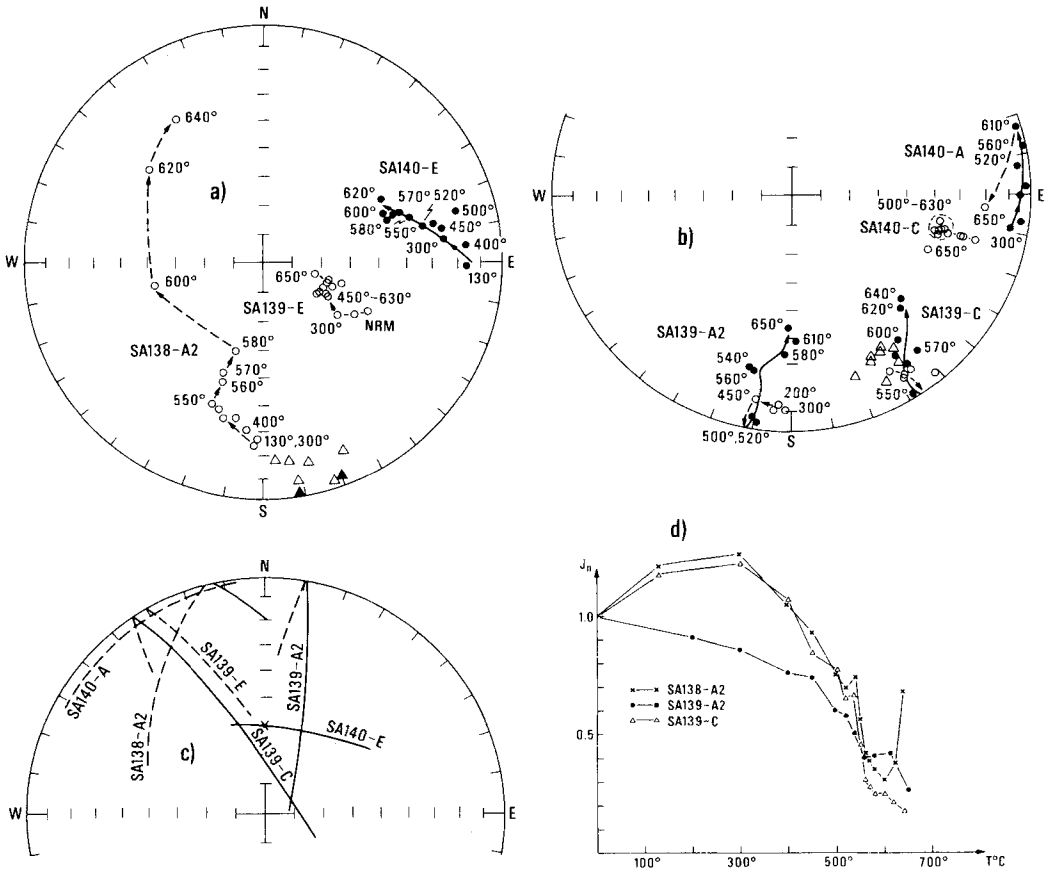


Figure 5. Thermal demagnetization results from site 24. Note the elongated distribution of the rather stable bulk magnetism and that remagnetization circles (c) tend to have both a shallow normal and an intermediately steep normal intersection 'point'. Diagram conventions as for Fig. 3. Further details in text.

both very low and very high blocking temperature phases. Specimen SA 132-B has a relatively strong zigzag pattern overprinting the general westward trend. By performing vector subtraction and by choosing temperature steps in such a way that the effect from these local irregularities is minimized, the direction of the erased vector becomes well defined and is in general agreement with the SSW directed magnetization. It is possible that the direction of the specimen is still changing above 600°C and the behaviour being analogous to that of specimen SA 129-B2 (Fig. 3c).

Site 24, Fig. 5, is another example where blocking temperatures above 600°C (probably associated with haematite) are carrying a relatively large portion of the natural magnetization. Again one notes a large within-site scatter of directions. Northerly directed trends accompanying increasing demagnetization are characteristic features but in general the stability spectra of the subcomponents are so closely overlapping that vector subtraction results are erratic and meaningless. For only two speciments, SA 138-A2 (Fig. 5a) and SA 139-C (Fig. 5b), are various directional estimates of the subtracted vector consistent, and again these fall in the SSE group. Specimen SA 138-A2 is the only one from this site for which a high temperature end point (normal) is being approached clearly. Fig. 5c shows extensions of the best-fitting great circle paths. These basically confirm the results from

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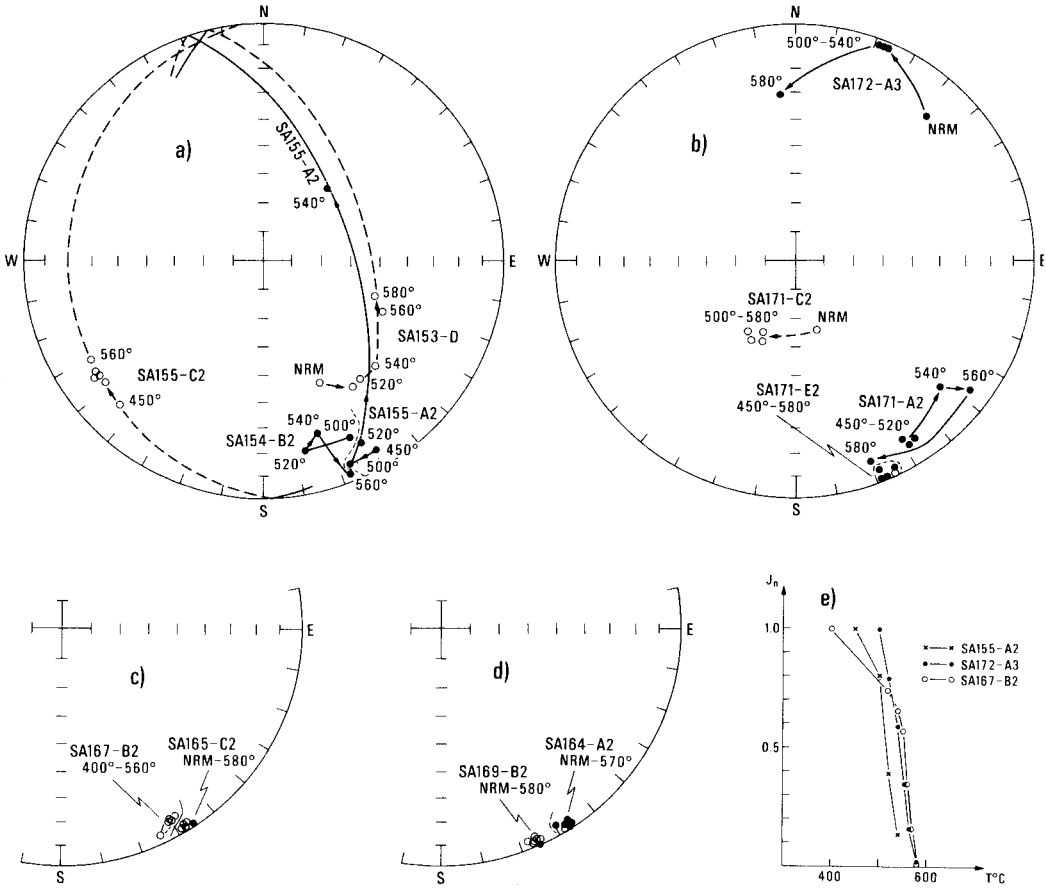


Figure 6. Thermal demagnetization results from four sites: (a) site 26, (b) site 31 and (c) and (d) sites 23 and 30. Further information in text and diagram details as for Fig. 3.

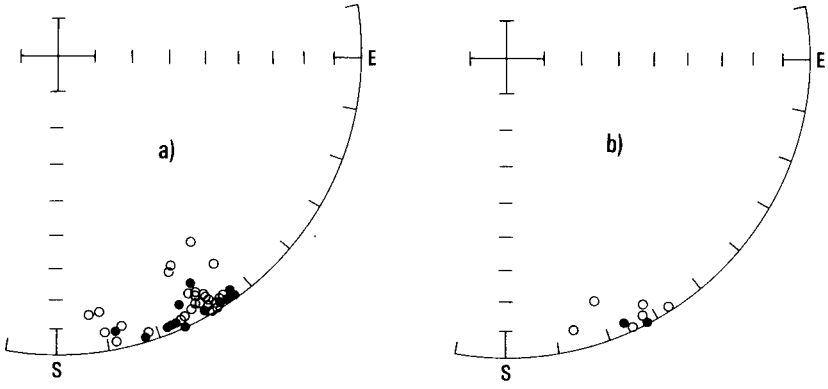


Figure 7. The distribution of the major magnetization group in the Sardinian ignimbrites: (a) specimen directions and (b) site mean directions.

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other sites suggesting a high temperature (haematite) direction anti-parallel to the characteristic 'magnetite' component (reversed). However, the trend of SA 139-A2 and SA 140-E differ from the rest in that they exhibit an intermediately steep normal, positive (down) intersection point. This supports the evidence from SA 129-B2 (Fig. 3c) that there is a certain subcomponent with high thermal stability magnetized in (or close to) the direction of the present axial dipole field.

Fig. 6 shows results from four sites: 23, 26, 30 and 31. Sites 23 and 30 are unique in this collection in that only a single, extremely well-defined, component of magnetization has been found. Examples from these two sites of tuffaceous material are given in Figs 6c and d.

Sites 26 and 31 (Figs 6a and b) confirm the general observation of this study that scattered remanence directions are often associated with swinging (mostly northerly) vectorial behaviour on increasing demagnetization. The convergence 'point' of remagnetization circles for site 26 is again in agreement with the evidence reported above of a shallow, north-north-westerly directed subcomponent of magnetization.

5 Discussion

With reference to the preceding chapters the magneto-mineralogy of these Sardinia volcanics indicates that there is a great variability in the state of oxidation and, except for sites 23 and 30, the full range of oxidation can be found even between separate specimens from a single hand sample. The magnetic properties of sites 23 and 30 are unique in that they apparently suggest a single palaeomagnetic component. As the natural magnetization of these sites does not seem to have been attacked by oxidation at all (in spite of the fact that the rocks carry numerous red oxidation 'spots') one is probably dealing with very small particles of pure magnetite having been exsolved within silicate grains. The location of the magnetization within silicate grains may be the reason why in this case it has been protected from modification by chemical remagnetization which has produced a quite complex multivectorial magnetization in most of the remaining sites. It is therefore argued that sites 23 and 30 possess the original cooling remanence (TRM or TCRM) acquired by these volcanics. The practically horizontal, SSE directed remanence, as exhibited by sites 23 and 30, is also found in the other sites; in localities with a strong magnetite component this direction is much more frequently extracted than in the more strongly oxidized sites. In the latter cases the SSE component was mostly obtained by application of vector subtraction.

The SSE directions of magnetization form the only consistent palaeomagnetic grouping in this investigation. All the relevant specimen results, the mean of stable end-point data and/or of vector subtracted directions, are listed in Table 1 along with formation mean data. Table 1 also contains 10 specimen results for which the bulk remanence, though fairly stable, has a very scattered distribution of directions. It is believed that these results represent unresolved multicomponent remanences and they are consequently not considered as actual palaeomagnetic field markers. These results are given in brackets. The remaining specimens, 23, not included in Table 1, either showed swinging directional behaviour (with no stable end-point or stable vector subtracted direction) or exhibited an unstable behaviour on thermal treatment.

The number of palaeomagnetic directions (specimens) in each site is inversely correlated with the degree of oxidation of the magnetic mineralogy. The overall mean direction of magnetization, $D \sim 153^\circ$, $I \sim -4^\circ$, is only slightly affected by the choice of unit vector (specimen or site). Least-squares fitting of the remagnetization circle for individual northerly trending directional paths and estimation of their best (normal) intersection point (Halls 1976) has been carried out for 12 specimens which exhibit well-defined northerly

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Table 1. Palaeomagnetic data from the Gallura ignimbrites, North Sardinia. See text for general details.

(a) Individual specimen direction (D, I) along with the temperature ranges of stable directions and/or the temperature intervals for which vector substration (vs) has been successful.

Site 21

SA 129-B1:	(218, +27);	300–580°
129-B2:	160, +13;	610, 630° + vs 200–520°
129-D1:	150, +12;	300–540° + vs 540–640°
130-B3:	145, −25;	610, 630°

Site 22

SA 132-B:	153, −20;	vs 300–600°
132-C1:	157, +3;	vs 300–560°
133-B3:	173, −14;	100–560°
133-C1:	171, −14;	130–580°
134-A1:	168, +5;	130–580°
134-A3:	170, −7;	NRM –590°
134-B1:	168, −3;	NRM –580°

Site 23

SA 135-A1:	144, +1;	NRM –540°
135-A2:	144, +1;	NRM –500°
135-B3:	146, −1;	NRM –540°
136-A1:	145, +2;	NRM –540°
136-B:	146, −1;	NRM –540°
137-A1:	147, −3;	NRM –540°
137-B2:	147, −2;	NRM –540°
137-C1:	146, −3;	NRM –550°
137-C2:	145, −3;	NRM –520°
164-A2:	146, +3;	NRM –570°
164-C2:	146, −2;	450–580°
165-B2:	149, −1;	200–570°
165-C2:	149, −3;	NRM –580°
166-A3:	162, −3;	NRM –520°
166-B2:	(130, +16);	400–560°

Site 24

SA 138-A2:	167, −8;	vs 130–640°
139-C:	152, −22;	vs 130–560°
139-E:	(113, −66);	450–650°
140-C:	(104, −37);	450–650°

Site 25

SA 151-B1:	163, +2;	500–580°
151-A3:	148, −5;	540–580°

Site 26

SA 153-A:	155, 0	vs 450–580°
154-B2:	160, +15;	500–560°
155-A2:	154, +9;	450–520°
155-B2:	(156, −48);	500–580°

Site 27

SA 156-B1:	(192, +6);	500–580°
159-B:	(106, +58);	500–580°

Site 28

SA 161-A2:	(209, −54);	500–560°

Site 30

SA 167-A2:	152, −5;	NRM –580°
167-B2:	150, −9;	400–560°
167-C1:	150, −6;	NRM –580°
167-C2:	148, −7;	NRM –580°
168-B2:	148, −8;	NRM –550°
169-A1:	154, −4;	NRM –570°
169-B2:	155, −3;	NRM –580°
169-C2:	151, −6;	NRM –560°
169-D1:	150, −10;	NRM –540°

Site 31

SA 170-A3:	151, −11;	450–580°
170-B2:	143, −15;	vs 500–560°
171-E2:	156, +3;	450–580°
171-A2:	160, +10;	580° + vs 520–560°
171-C2:	(207, −61);	500–580°
172-A3:	(023, +3);	500–540°

(b) Mean formation data; directions in brackets not included.

N	D_m	I_m	κ	α_{95}	
44	153.2	−3.6	49	3.1	distribution of specimen directions
8	154.9	−3.9	99	5.6	distribution of site mean directions

N: number of unit vectors; D_m: mean declination; I_m: mean inclination;

κ: precision parameter, $\frac{N-1}{N-R}$, where R is length of resultant vector;

α_{95}: radius in the 95 per cent circle of confidence.

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paths as well as tending to have a common intersection point. This convergence point is: $D = 338°$, $I = 0°$, i.e. closely anti-parallel to the SSE group. The northerly directional trends are more pronounced in the strongly oxidized specimens, i.e. those with a high haematite content, and therefore the NNW direction is likely to represent a field component impressed through remagnetization rather than being a result of partial self-reversal. The result suggests that the dominant remagnetization processes occurred in roughly the same dipole field in which the original magnetization was acquired. However, the experimental data suggest that the field was consistently reversed at the time the volcanics originated (*cf.* the magnetite remanences) while a normal (or alternating) polarity characterized the field during the subsequent oxidation and partial remagnetization. The primary tectonic situation and the lengthy magnetization history (impressing time-average palaeomagnetic vectors even in single specimens) suggest that the available data may serve as an adequate basis for palaeomagnetic and tectonic conclusions for Sardinia.

If one considers Sardinia as being part of the European plate in Permian time there is a small but statistically significant difference of 8° between the overall inclination for these volcanics and the inferred relative inclination for the Permian of Europe. This inclination difference may have various explanations but when seen in conjunction with the polarity pattern in the earliest history of these rocks (outlined above) the following suggestion seems appropriate: the rocks were formed at the end of the stable Kiaman geomagnetic epoch of steady reversed polarity (~ 300–230 Myr) becoming subsequently partially remagnetized in a geomagnetic field characterized both by fluctuating polarity and a gradual axis change towards the Triassic position. Thus, the total palaeomagnetic evidence is in favour of a late Permian origin for these volcanics which is in line with the proposition of Zijderveld (1976, private communication) that the Sardinian ignimbrites and the late Permian volcanics in the Esterel region of southern France (Zijderveld 1975, unpublished) may be roughly time equivalent formations. The slightly higher overall palaeomagnetic inclination of the Esterel rocks compared to that of the Sardinian rocks possibly indicates that the Esterel rocks underwent a slower and longer-lasting magnetization history, so acquiring a stronger Permo–Triassic field component than the Sardinian rocks.

The acceptance of Sardinia as originally part of the European plate leads to the inevitable conclusion than an anticlockwise rotation of the island, of around 45°, must have taken place in post-Palaeozoic time. Whether also Corsica participated in such a rotation or not is still a matter of dispute. Nairn & Westphal (1968) and Westphal *et al.* (1976) on the basis of late Palaeozoic palaeomagnetism claim to have found evidence for an anticlockwise rotation of *c.* 30° while Storetvedt & Petersen (1976) through detailed demagnetization studies of Corsican rocks of similar age found no support for this idea. However, the magnetization of the late Palaeozoic rocks in Corsica is clearly complex so further detailed work is necessary before this issue can be settled. Whatever the final result will be, the suggested major fault running through the Strait of Bonifacio (Bayer, le Mouël & le Pichon 1973) may indicate a crustal decoupling between Corsica and Sardinia, so clearly there are no obvious reasons for Corsica to have been affected by the processes that seem to have caused rotation of Sardinia.

The favoured explanation of the postulated rotation of Corsica and Sardinia has been that these islands were originally situated near the southern coast of France, the subsequent rotation(s) having been accomplished through anticlockwise movements around Euler poles in the Gulf of Genoa region (Nairn & Westphal 1968; Alvarez 1972; Westphal *et al.* 1976). Such movements are consistent with the Ligurian Sea basin having been formed as a consequence of sea-floor spreading but the problem with this is that the recorded anomaly pattern in the Ligurian Sea does not conform to the suggested type of rotation (Storetvedt 1973b). Furthermore, this anomaly pattern is not confined to the deep-sea region but continues its

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trend 'onto' the Sardinia and Corsica blocks (Storetvedt 1973b). This seems incompatible with a spreading origin for these magnetic lineaments. The anomaly pattern in the West Mediterranean has two principal trends: NE–ENE and SW–WSW. These systems are most likely a surface expression of shear fractures (with magmatic infill) having formed as a result of a northeasterly movement of Africa relative to Europe so dividing the West Mediterranean into a mosaic of independently moving crustal fragments. The anticlockwise rotation of Sardinia is therefore seen as an *in situ* rotation within the framework of a tectonically subdivided crust.

6 Conclusions

The conclusions are mainly based on palaeomagnetic information from 44 specimens (eight sites) out of a total of 77 tested specimens from 11 sites. From the experimental data and considerations outlined above we propose the following sequence of events in the magnetization history of the Gallura ignimbrites:

(a) The rocks were laid down in late Permian time at the end of the reversed Kiaman geomagnetic epoch. The dipole axis had probably already started to change towards a Triassic position but the polarity was still consistently reversed. At this time Sardinia is thought to have formed part of the stable European plate and to have had a palaeomagnetic mean direction of $D \sim 200°, I \sim 0°$.

(b) After cooling, the rocks were exposed to surface conditions in a tropical environment in late Permian or possibly Triassic time when they were oxidized and partially remagnetized. The extent of these secondary, low-temperature processes varies greatly, from very little or no alteration to practically complete alteration of the primary magnetization. The most important remagnetization field was of normal polarity and approximately anti-parallel

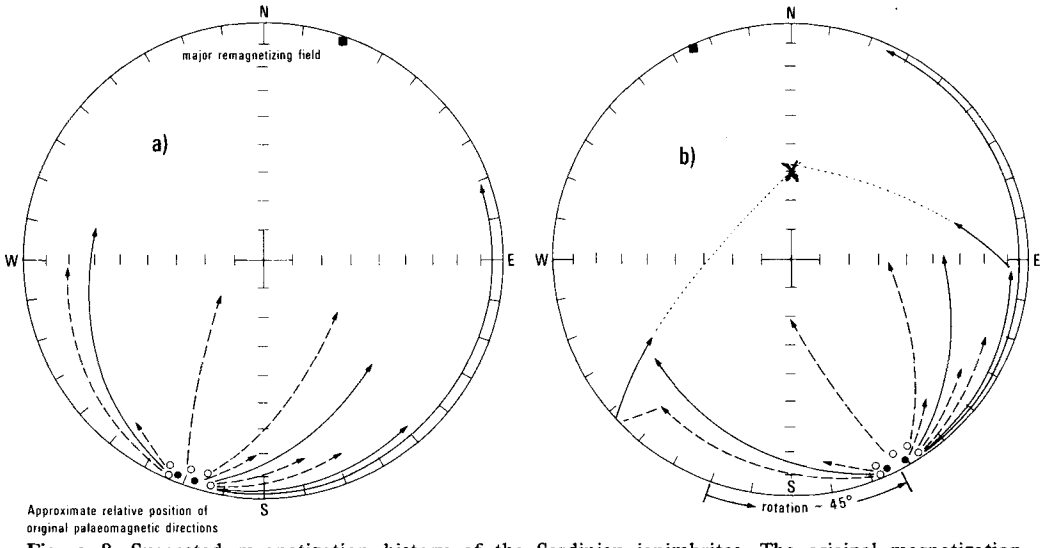


Figure 8. Suggested magnetization history of the Sardinian ignimbrites. The original magnetization (diagram a) was acquired in a practically horizontal, south-southwesterly field, which is thought to represent the Upper Permian just before the close of the Kiaman geomagnetic epoch. Subsequent remagnetization basically along an approximately anti-parallel field direction, led to a scattered distribution of the resultant magnetization in most sites (the end of the arrows may indicate the degree of maximum directional spread). This late Permian–Triassic remagnetization event was followed by a 45° anticlockwise rotation of Sardinia (diagram b) after which a certain magnetization component (probably of chemical origin) has been impressed in the direction of the present axial dipole field. Diagram conventions as for Fig. 3.

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to the ambient field direction at the time of cooling. Superposition of the secondary normal component(s) led to a general spreadout of the resultant magnetization vectors with a preference for easterly trends as illustrated in Fig. 8a. To some extent secondary components of magnetization were also impressed during periods of reversed field polarity. On the whole, the more strongly oxidized material ended up with a complex, fairly scattered distribution of resultant magnetization directions. On the other hand, rocks which escaped this early low-temperature chemical alteration of their primary oxides have retained the original single-component remanence.

(c) At some later date, probably in the Tertiary, Sardinia underwent an anticlockwise rotation of about 45°, establishing a new relative position of the total magnetization structure as illustrated in Fig. 8b. This rotation of Sardinia is seen as an *in situ* motion, resulting from crustal deformation caused by differential movement between Africa and Europe. Subsequent to the rotation, a minor chemical remanence with blocking temperature $\geqslant 600°C$, has been acquired along the direction of the present axial dipole field.

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